

Sound waves and waves for detection and exploration

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 6 starred (★) questions first.

	QUESTION	ANSWER	✓
1	A sound wave passes into a different medium. Which stays the same: speed, frequency or wavelength?	Frequency	☐
2	State what happens to the speed of sound when it passes from air into a solid.	It increases.	☐
3	State what happens to the loudness of a sound passing from air into a solid, and why.	It decreases: some sound is reflected or absorbed at the boundary.	☐
4	Name the part of the human ear that detects sound waves.	The eardrum	☐
5	Name the part of the ear that sends electrical signals to the brain.	The cochlea	☐
6 ★	State the range of human hearing.	20 Hz to 20 kHz	☐
7 ★	Name sound with a frequency above the range of human hearing.	Ultrasound	☐
8	What is echolocation?	Using reflected sound (echoes) to locate an object.	☐
9	Give two uses of ultrasound imaging.	Medical scans (e.g. of an unborn baby) and finding flaws in materials	☐
10	What are seismic waves?	Waves produced by earthquakes.	☐
11	Name the two types of seismic waves.	P-waves and S-waves	☐
12 ★	What type of wave are P-waves?	Longitudinal	☐

QUESTION		ANSWER	✓
13	Name the states of matter that P-waves travel through.	Solids and liquids	☐
14 ★	What type of wave are S-waves?	Transverse	☐
15	Name the state of matter that S-waves travel through.	Solids only	☐
16	Describe how a sound wave travels through a solid.	The particles of the solid vibrate and pass the vibrations on.	☐
17	Explain why human hearing is limited to a range of frequencies.	Sound makes the eardrum vibrate only over a limited range of frequencies.	☐
18	State what happens to ultrasound at a boundary between two media.	It is partially reflected.	☐
19 ★	Describe how ultrasound is used to find the distance to a boundary.	Time the reflection: distance = speed × time ÷ 2.	☐
20	State what echo sounding is used for.	Detecting objects in deep water and measuring water depth.	☐
21	State what seismic waves have provided evidence for.	The structure and size of the Earth's core.	☐
22 ★	Explain how S-waves provide evidence that part of the Earth's core is liquid.	S-waves cannot travel through liquid and are not detected on the far side of the Earth.	☐